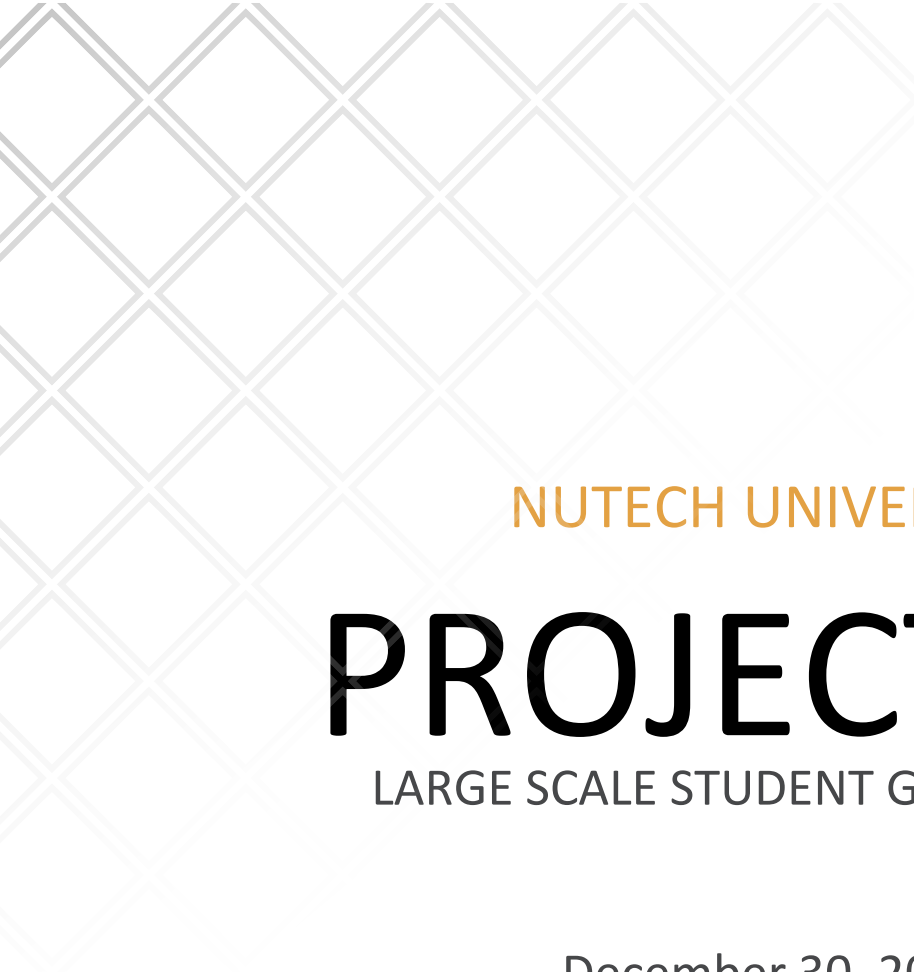


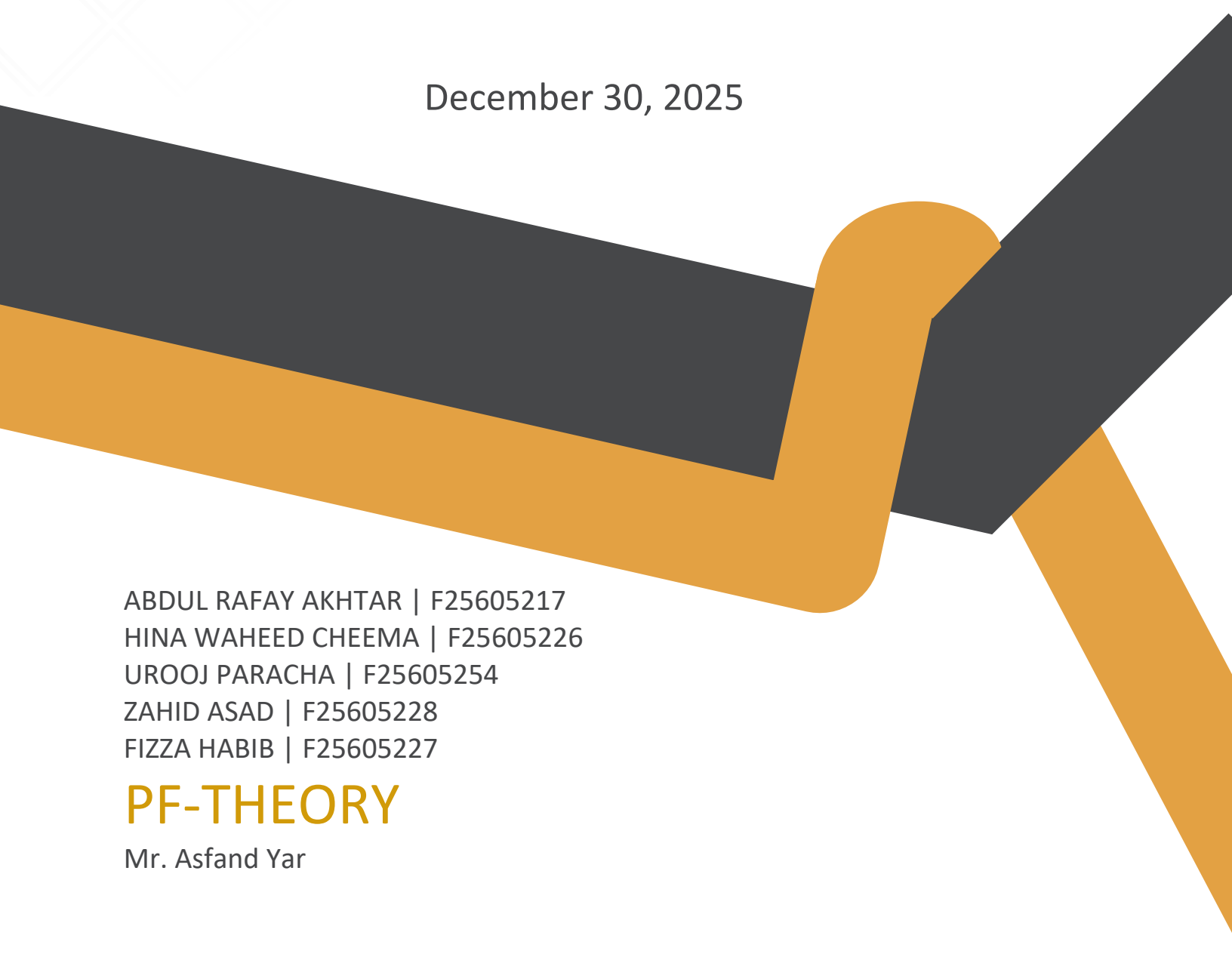


NUTECH UNIVERSITY

PROJECT C++

LARGE SCALE STUDENT GRADING SYSTEM

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PF-THEORY

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Computer Science Department

Semester fall – 2025

Program: Computer Science

Course: PF Theory

Course Code: CS120

Submitted To:

Mr. Asfand yar



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PROJECT C++

Project Title: large scale student grading system

1. Introduction

Hello! This is our first semester programming project. I have created a complete Student Grading System using C++. This project combines all the basic programming ideas we learned in my first semester. It shows how to solve real problems using computer programming.

The system helps teachers manage student grades for both schools and universities. It is a practical tool that can be used in real educational institutions. we built this project to show that we understand how to write complete, working programs.

2. Objectives of the Project

My main goals for this project were:

- **Apply what we learned:** Use all the C++ concepts from class in one big project.
- **Solve a real problem:** Create something useful for schools and colleges.
- **Improve my thinking skills:** Learn to break big problems into small steps.
- **Build confidence:** Show myself I can make complete programs.
- **Practice coding:** Write a lot of code to become better at programming.

we wanted to prove that we can take classroom knowledge and use it to make something that actually works.

3. Programming Concepts Used

we used almost every concept we learned this semester:

Basic Concepts:

- **Input/Output:** cin and cout for user interaction.
- **Data Types:** int, double, string, bool for different kinds of data.
- **Operators:** Arithmetic (+, -, *, /), comparison (<, >, ==).

Control Structures:

- **Conditional Statements:** if, else if, else, switch for decisions.
- **Loops:** for, while, do-while for repeating tasks.

Advanced Concepts:

- **Arrays:** To store lists of students and subjects.
- **Functions:** 15+ functions to organize the code.
- **Structures:** struct to create custom data types for students and teachers.
- **File Handling:** Save and load data from text files.
- **Input Validation:** Check if user enters correct information.

4. System Description

This is what my program does:

Main Features:

1. Two User Types:

- **Teacher Portal:** Requires login (ID: 101, Password: asfand(#123)).
- **Student Portal:** View reports without login.

2. Two Education Systems:

- **School/College:** For classes 1 to 12
- **University:** For degree programs (BS, MS)

1. What Teachers Can Do:

- Add new students
- Enter student marks
- Calculate results
- View and print report cards
- Search for students
- Save and load data

2. What Students Can Do:

- View their own report cards by entering roll number.

Menu System:

The program has simple menus that guide the user:

MAIN MENU → TEACHER LOGIN → MAIN TEACHER MENU → Choose option (1-8)

Everything is organized into clear sections so users don't get confused.

5. Algorithm

Here are the steps my program follows:

Step 1: Start

- Initialize teacher account.
- Try to load saved student data.

Step 2: Show Main Menu

Choose:

1. Teacher Portal
2. Student Portal
3. Exit

Step 3: If Teacher Chosen

1. Ask for Teacher ID and Password

2. If correct → Show Teacher Menu
3. If wrong → Show error and go back

Step 4: Teacher Menu Options

1. Enter Student Details

- Get roll number, name, class
- Set up subjects based on class

2. Enter Student Marks

- Find student by roll number
- Enter marks for each subject

3. Calculate Result

- Find student
- Calculate percentage/GPA
- Determine grade

4. Display Report

- Show beautiful report card
- Print all subjects and final result

5. Search Student

- Find by roll number or name

6. Load Data

- Read saved students from file

7. Save Data

- Write all students to file

8. Exit

- Save and return to main menu

Step 5: If Student Portal Chosen

- Ask for roll number
- Display report if result is calculated

Step 6: Repeat

- Keep showing menus until user chooses Exit
- Save data when exiting

6. Implementation Details

How I Organized the Code:

I created **20 separate functions** - each does one specific job:

1. Main Functions:

- EnterStudentDetails() -Add new student
- EnterStudentMarks() - Enter marks
- CalculateResult() - Calculate grades
- DisplayReport() - Show report card

2. Helper Functions:

- findStudentIndex() - Find student by roll number
- marksToGradePoints() - Convert marks to GPA
- getGradeLetter() - Get letter grade
- getSchoolGrade() - Get school grade

3. System Functions:

- teacherLogin() - Check teacher password
- loadAllData() - Read from file
- saveData() - Write to file

Data Structures:

I created three custom types using struct:

```
struct Student {  
    int roll_number;  
    string name;  
    // ... 15+ more fields ...  
    Subject subjects[20]; // Each student has subjects  
};
```

```
struct Subject {  
    string name;  
    int max_marks;  
    double obtained_marks;  
    // ... and more  
};
```

```
struct Teacher {  
    int id;  
    string name;  
    string password;  
};
```

Important Arrays:

```
Student records[100];    // Can store 100 students  
Teacher teachers[1];     // One teacher account
```


Special Features I Added:

1. Different Subjects for Each Class:

- **Class 1-2:** English, Urdu, Maths, GK, Islamiyat, Drawing
- **Class 9-10:** Different for Bio vs Computer Science
- **University:** Full semester-wise courses for BS programs

2. Different Grading Systems:

- **Schools:** Percentage-based (A+, A, B, etc.)
- **Universities:** GPA-based (4.0 scale)

3. University Components:

- Quiz (10%), Assignment (10%), PBL (10%), Mid (20%), Final (50%)

4. Professional Report Formatting:

- Uses setw() and formatting for nice tables
- Shows all details in organized way

7. Sample Output

OUTPUT SCREENSHOT:

```
C:\Final PF Project C++\Large × + v
Teacher account initialized: Asfand Yar (ID: 101)

===== Loading Student Data =====
INFO: No saved data found. Starting fresh.

=====
STUDENT GRADING SYSTEM – USER SELECTION

=====
1. Teacher Portal
2. Student Portal
3. Exit
Enter your choice: |
```

```
=====Login Menu=====
```

```
Enter Teacher ID: 101
```

```
Enter Password: asfand(#123)
```

```
*****
```

```
*** LOGIN SUCCESSFUL! Welcome: Asfand Yar ***
```

```
*****
```

```
=====
                        STUDENT GRADING SYSTEM - TEACHER PORTAL
=====
```

1. Enter Student Details
2. Enter Student Marks
3. Calculate Result
4. Display Report
5. Search Student Record
6. Load All Student Data
7. Save All Student Data
8. Exit

```
Enter Choice (1-8): |
```

```
Enter new Student roll number (must be unique): 217
```

```
Enter Student Name: ABDUL RAFAY AKHTAR
```

```
Enter Class (1-12) or 33 for university: 33
```

```
----- Enter University Data Collection (Nutech University) -----
```

```
Select Degree Level:
```

```
1. BS (Bachelor of Science)
```

```
2. MS (Master of Science)
```

```
Enter Degree Choice (1-2): 1
```

```
Select Department:
```

```
1. Computing & IT Department
```

```
2. Engineering Department
```

```
Enter Department Choice (1-2): 1
```

```
Select Program:
```

```
1. BS Computer Science
```

```
2. BS Software Engineering
```

```
Enter Choice (1-2): |
```

```
Select Program:
1. BS Computer Science
2. BS Software Engineering
Enter Choice (1-2): 1

Enter Current Semester (1-8): 1

--- Loading Curriculum for BS Computer Science Semester 1 ---

=====
Enrollment Complete!
Student Name: ABDUL RAFAY AKHTAR
Roll Number: 217
Total Enrolled Students: 1
=====
Press Enter to continue...|
```

```
===== Enter Student Marks =====
Enter Student Roll number to enter marks for: 217

--- Student Found: ABDUL RAFAY AKHTAR ---
University: BS Computer Science (Semester 1)

--- University Component Marks Entry ---
Component Weightage: Quiz(10%), Assignment(10%), PBL(10%), Mid(20%), Final(50%)

Subject: Programming Fundamentals (3 Cr) (3 Credit Hours)
Enter Quiz Marks (0-10): 10
Enter Assignment Marks (0-10): 7
Enter PBL Marks (0-10): 6
Enter Midterm Marks (0-20): 21
ERROR: Invalid marks. Please enter between 0 and 20
Enter Midterm Marks (0-20): 17
Enter Final Exam Marks (0-50): 50
-> Total marks for Programming Fundamentals (3 Cr): 90/100
```

```
===== Calculate Student Result =====
Enter Student Roll number to calculate result for: 217
Processing result for University student (GPA system)...

=====
RESULT CALCULATED SUCCESSFULLY!
Final CGPA: 3.70/4.0
Overall Grade: Distinction (A+)
Use Option 4 (Display Report) to view the full result card.
=====
Press Enter to continue...|
```

```

===== Display Student Report =====
Enter Student Roll number to display report for: 217

=====
STUDENT ACADEMIC REPORT CARD
=====
Institution:      Nutech University
Head:            Lt. Gen. Moazzam Ejaz (Rector)
Class Teacher:   Asfand Yar
Academic Year:   2024-2025

-----
Student Name:    ABDUL RAFAY AKHTAR
Roll Number:     217
Program:         BS Computer Science
Department:      Computing & IT Department
Semester:        1
Degree Level:    BS

=====
SUBJECT WISE MARKS
=====
COURSE                                CR.HRS    MARKS    GRADE    COMPONENTS
-----
Programming Fundamentals (3 Cr)        3         90      A+ Q:10 A:7 M:17 F:50
Calculus and Analytical Geometry (3 Cr) 3         75      B+ Q:6 A:5 M:14 F:43
Functional English (3 Cr)              3         85      A+ Q:6 A:7 M:17 F:47
Discrete Structures (3 Cr)             3         78      B+ Q:4 A:6 M:18 F:43
ICT (2 Cr)                             2         83      A Q:8 A:7 M:17 F:43
ICT LAB (1 Cr)                        1         94      A+ Q:9 A:8 M:20 F:50
PF LAB (1 Cr)                         1        100      A+ Q:10 A:10 M:20 F:50

=====
FINAL RESULT
=====
Total Credit Hours:      7.0
CGPA:                    3.70/4.0
Percentage Equivalent:   86.43%
Overall Status:          Distinction (A+)

=====
Remarks: PASS - Eligible for next semester

```

8. Conclusion

This project was a **big learning experience** for us. It took us from knowing individual C++ concepts to being able to combine them into a complete, working program.

What this project proves:

1. we understand C++ fundamentals
2. we can apply theory to practical problems
3. we can organize and write large programs
4. we can solve real-world problems with code

Most importantly, this project showed us that programming is not just about writing code, it's about solving problems, helping people, and creating useful tools.

We are proud of what we built, and we are excited to learn more and make even better programs in the future!

9. Future Enhancements

If I had more time or skills, I would add:

1. **More teacher accounts:** Allow multiple teachers.
2. **Password security:** Hide password when typing.
3. **Graphical Interface:** Use graphics instead of text menus.
4. **Database:** Use real database instead of text files.
5. **Network feature:** Multiple computers can use it.
6. **Print reports:** Directly print to printer.
7. **AI features:** Predict student performance.
8. **Parent portal:** Parents can check grades.

THANK YOU!